SPRING 2026 - CMPE362 - HOMEWORK 4
Filtering

The aim of this homework is to design and apply filters using MATLAB. You are given an audio file `cafe_sample.wav`.

Names of some of the filter design functions in MATLAB (you are not limited to these).

`fir1, fir2, firls, firpm, firceqrip, firgr, kaiserord, firrcos, designfilt, butter, cheby1, cheby2, ellip, bessel, yulewalk, iirnotch, iirpeak, invfreqz, invfreqs, tf2sos, zp2tf, sosfilt, dfilt, fdesign, design`

To the best of your abilities attempt to separate human speech in the audio with the music in the background using filters. Generate two audio files:

- `speech_filtered.wav`: Human speech is filtered out, background audio and music is audible.
- `music_filtered.wav`: Background noise and music is filtered out, human speech is audible.

Note: You will most likely not be able to separate the two parts very clearly. Do the best you can. A little bit of audio quality degradation is to be expected

Optional Part (Bonus):

There’s a text message (periodically) hidden in the given audio file. Can you decode it?

Submission

Upload:

- All MATLAB scripts (.m files)
- All audio files (.wav files)
- A presentation (.pdf)

Presentation Format

- (1 slide) Title Page (your name, student ID, etc.)
- (1-2 slides) Spectrograms for all three audio files `cafe_sample.wav, speech_filtered.wav, music_filtered.wav`
- (1 slide) Mention the filters you have used, frequency ranges that you have targeted, your reasoning and process.
- (1 slide) If you decoded the message in the bonus section, explain your process and also the message.

LLM Disclaimer

You are allowed (even encouraged) to use LLMs while doing this homework. However:

Please do not copy large blocks of generated explanations into your slides.
Try to interpret what you observe and write short, clear comments in your own words.

The grading will be done charitably, clarity and engagement matter more than perfection.